Broken-Stitch Detection under Augmentation Leakage: A Source-Lineage Audit and Cross-Domain Evaluation

***Abstract*—Image augmentation can create thousands of stitch-defect examples without adding independent source photographs. We audit an unpublished garment-inspection collection previously associated with 94.91% reported VGG-16 validation accuracy. Its recovered ZIP contains 7,365 images (7,364 readable). Removing augmentation UUIDs identifies 182 candidate image families, *all* represented in multiple original train/validation/test folders. Filename structure and visual matches further suggest that 151 digitally edited faulty families descend from only 31 normal-photo candidates. In an exploratory five-fold development-only comparison with identical frozen MobileNetV3-Small features and classifier settings, an image-random protocol scores 0.871 macro-F1 while inferred-photo-parent-grouped evaluation scores 0.688, an 18.35-point protocol difference. On a separate 21/5/5 inferred-photo-parent split, a fixed ImageNet-pretrained MobileNetV3-Small feature extractor and balanced logistic head obtain 80.20% accuracy, 0.799 macro-F1 and 74.85% faulty recall on 1,086 initially held-out augmented images from five candidate source photographs. Separately, an *earlier*, differently trained MobileNet reaches 0.635 macro-F1 and only 57.79% broken-stitch recall on 3,637 real StitchingNet normal/broken images, with no external tuning. The two tests involve *different* frozen checkpoints and must not be conflated. These observations expose a gap between nominal validation scores and credible defect-screening evidence. Since original photographed-source lineage has not been independently authenticated, our grouped results remain exploratory rather than validated factory generalization.**

***Index Terms*—broken stitch detection, augmentation leakage, source-grouped evaluation, MobileNetV3, synthetic-to-real transfer, reproducibility**

I. Introduction

A broken seam stitch is a small interruption in a repeated, often low-contrast thread pattern. Its appearance changes with fabric, camera distance, lighting and thread colour. Deep classifiers and pretrained features are appealing in garment inspection, but defect collections are frequently enlarged by digital editing and geometrical augmentation. The corresponding image count is then a poor proxy for the number of independent garment or photographic sources. Grouping transformations by the wrong unit can cause related images to occur in training and testing, making apparent generalization misleading.

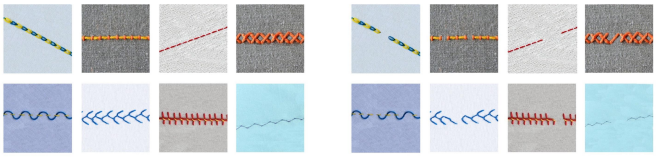
An unpublished 2022 thesis evaluated activation functions, filters, a CNN, VGG-16 and neural-network ensembles for correct-versus-faulty stitches, reporting a final 94.91% *validation* accuracy. Its original source code and trained checkpoint are unavailable; the present investigation is a new, non-identical PyTorch reconstruction. The contribution is a *case-specific, auditable evaluation protocol*, not a novel network. We (i) detect systematic cross-split augmentation-family contamination despite zero exact-byte duplicate groups, (ii) reconstruct and explicitly qualify potential normal-to-Photoshop-defect photo ancestry, and (iii) separate the results of frozen models on inferred-parent internal data and authentic external sewing defects. A development-only controlled partition diagnostic (described below) directly probes split-policy sensitivity without reusing the previously inspected holdout.

Three questions guide the study: *RQ1*: did the original data folders divide augmentation families across partitions? *RQ2*: what happens when normal images and their likely Photoshop-faulty derivatives share a candidate photo-parent group? *RQ3*: how do strictly frozen models trained on this small synthetic collection perform on genuine sewing-stitch data in another acquisition domain? Importantly, the new inferred-photo-grouped model and the first externally tested model are different runs. The underlying photo-parent relationships remain inferred rather than independently authenticated.

II. Related Work

Early textile inspection employed handcrafted texture features and image analysis [1]. The general leakage problem is documented by Kapoor and Narayanan [7]; Yagis *et al.* showed how slice-level rather than patient-level partitioning alters apparent image-classification performance [8]. Gerz and Jelali [9] detail the threat of augment-then-split evaluation. These works establish that leakage itself is *not* new; our contribution is forensic documentation and controlled evaluation of this particular industrial-stitch archive. Kim *et al.* combined CNN feature maps and image processing for sewing broken-stitch detection [2]; Chowdhury *et al.* compared CNN, VGG-16 and VGG-19 on 500 stitch images [3]. These precedents mean that adding a pretrained VGG model is not, by itself, a novel contribution. The StitchingNet publication [4] introduced 14,565 sewing-stitch images spanning normal and ten defect types across 11 fabric/thread configurations, enabling more demanding tests of transfer. More recent sewing-image research studies anomaly-based inference [5] and localized segmentation [6]. The latter derives from StitchingNet, and hence cannot serve as an *independent second* external test after that source has been evaluated.

Compared with these studies, our narrower binary task probes a data-accounting question: whether nominally large



(a) Historical normal example (b) Digitally constructed defect

Fig. 1. Historical thesis examples; (b) must not be interpreted as a naturally occurring production defect. Image reuse rights require author review.

augmented datasets really support independent-source accuracy claims. Differences in camera, defect definition, splits, task and score preclude direct performance rankings between published systems.

III. Data and Audit Procedure

A. Historical collection and recovered archive

The thesis describes 31 correct images and roughly 150 digitally modified faulty images, with some images obtained from garment sites and the internet. Because photo capture identity and editing history are incomplete, these figures are *not* 181 verified independent photographs. The recovered 2,146,784,353-byte ZIP contains 7,365 image records (2,734 labelled `actual`, 4,631 `faulty`), of which 7,364 decode at 400×400 pixels; one normal PNG is unreadable. The 151 named faulty candidate families do not exactly reconcile with the approximately 150 faults described in the thesis. Their publication-level source ownership, editor provenance and garment identity have not been established, so the archive is treated as a forensic case study rather than a licensable independent photographed-source benchmark.

B. Two levels of source lineage

Each filename includes a base identifier and a UUID-like augmentation suffix. After stripping the suffix, 182 candidate image families remain: 31 `actual_original_XX` and 151 `faulty_original_XXN`. Each of the 182 family identifiers occurs in at least two original folder partitions. Accordingly, *every one* of the 7,365 images belongs to an image family spread across original train/validation/test folders. There are zero identical-byte SHA-256 image groups, demonstrating why exact-byte deduplication alone does not detect transformed derivatives.

At a second, more conservative level, the filename pattern `faulty_original_XXN` points to `actual_original_XX`. All 151 faulty families syntactically map to one of the 31 normal photo candidates; automated visual similarity ranks support multiple matching pairs. This is evidence for an *inferred* 31-parent graph, not proof of the original camera photographs or Photoshop edit chains. A single physical seam may also have been photographed several times, making even 31 an upper bound on reliably distinct physical sources without manual verification.

C. Evaluation partitions and provenance controls

The original train/validation/test folder membership is rejected for independent evaluation. Rather than relying on the number of JPEG/PNG records, the relevant unit of independence is the source photograph (and potentially the physical garment). A fixed random seed (20261018) groups the 31 inferred photo parents into 21 training, five validation and five internal holdout parent IDs, excluding one corrupt image. The resulting 7,364 valid images are: 5,019 training (1,878 normal, 3,141 faulty), 1,259 validation (433/826) and 1,086 holdout (422/664). No inferred photo-parent key crosses these new partitions. The parent IDs represent *candidate lineages*, not 31 confirmed independent garments. In this retrospective study, all test records are augmented or Photoshop-edited examples; the test does not contain a new, independently photographed real-defect collection. Nonetheless, group independence is conditional on the unverified filename-to-real-photo relation. Checkpoints, ZIP digests and partition hashes are recorded before the holdout is unsealed. The holdout workflow verifies the frozen checkpoint SHA-256, original archive checksum and manifest fingerprint, and refuses to refit models or alter the decision threshold.

IV. Models and Experiments

A. Reconstructed scratch CNN

The new PyTorch SmallCNN reconstructs an experimental model *family*, not the exact 2022 TensorFlow/Keras weights. An exploratory initial run used the *older* 182-family grouping, input resolution 100×100, class-weighted cross entropy, Adam (10^{-3}), batch size 32, train-time augmentation and selection by validation macro-F1 over three completed epochs. Its 1,077-image candidate-family holdout returned 57.38% accuracy and zero recall for normal stitches. This failure is retained as a software-verified diagnostic, not a comparable result for the subsequent 31-photo grouping.

B. ImageNet-pretrained MobileNetV3-Small

Two *distinct*, fixed MobileNetV3-Small experiments [10] use ImageNet1K-V1 pretrained *frozen* convolutional features with 224×224 RGB input, ImageNet normalization, global spatial averaging and a standardized balanced logistic regression classifier ($C = 1$). The class decision threshold is 0.5; no hyperparameter sweep was carried out. The first run used the older 182-family split (5,167 training and 1,120 validation images). Its validation macro-F1 was 0.648; no holdout result from that run is claimed in this paper. The second used the inferred 31-photo-parent grouping (5,019 training, 1,259 validation); this second model alone is evaluated on the first-time 1,086-image internal parent holdout. They have different fitted heads and manifests, and must not be treated as the same model.

C. Development-only CV split-policy diagnostic

A supplementary five-fold experiment uses *only the original 26 inferred photo-parent groups assigned to development* (5,019 train and 1,259 validation records, totaling 6,278). The

five parents previously unsealed for the internal holdout and all StitchingNet images are *excluded*. On a single set of frozen ImageNet MobileNetV3-Small feature vectors, we compare (a) deliberately leakage-prone image-stratified cross-validation and (b) inferred-photo-parent-grouped cross-validation. For every fold, standardization and a fixed balanced logistic head ($C = 1$, threshold 0.5) are fitted *only* on fold-training records. The input images, architecture, folds count and hyperparameters are held fixed across protocols; only partition policy differs. Since group-wise composition and image-level ease also differ, this is a *protocol-sensitivity diagnostic*, not a causal estimate of inflation in the historical 94.91% score. It was conceived after the first holdout was inspected, so it is additionally exploratory and is not called a new independent test.

D. First external test and separation of evidence

Before inspecting the real StitchingNet test metrics, the *older* 182-family-trained MobileNet and SmallCNN checkpoints and threshold were frozen. The official StitchingNet archive checksum was checked; for the binary task, only normal class 00 ($n = 2{,}436$) and broken-stitch class 02 ($n = 1{,}201$) were retained. The nine other defect classes were excluded. No external training, calibration or score-based selection was performed. Byte-identical overlap with the original ZIP was not detected, although transformed/semantic overlap cannot be excluded from hash checking alone. StitchingNet was examined *once* in this protocol; later tuning cannot reuse it as an untouched external benchmark. The *new* 31-photo-parent MobileNet has *never* been evaluated on StitchingNet.

V. Results and Discussion

A. Original-split audit

The original archive's systematic cross-partition sharing of *all* 182 augmentation families invalidates any assertion that its existing folders measure independent-image-family generalization. This finding does not prove the historical 94.91% validation result used exactly the recovered folders, because its complete training scripts and run manifests are unavailable. The historical accuracy is therefore a source-reported number, not a reproduced result that can be directly subtracted from our new measurements.

B. Controlled development-only partition diagnostic

The same 6,278 development images from 26 inferred source-parent candidates were subjected to two fixed five-fold protocols. Under image-random stratified folds, variants from *all 26* candidate parents occurred in both training and validation in each fold; under inferred-parent-grouped folds, this count was *zero* in every fold. All frozen feature extraction and classifier settings were unchanged. Table I reports pooled out-of-fold (OOF) metrics: macro-F1 decreases from 0.8714 to 0.6879 (18.35 percentage points), and balanced accuracy from 87.77% to 69.01%. This is a direct empirical measure of *protocol sensitivity in this archive*, not proof that the unavailable historical 2022 model was inflated by 18.35 points. Different group composition and the inferred (not authenticated) parent mapping still confound any causal interpretation. In addition, this retrospective diagnostic was conducted after prior holdout scores were examined and is *not* an independent test.

TABLE I
SAME PRETRAINED BACKBONE AND CLASSIFIER; 5-FOLD DEVELOPMENT-ONLY CV ($N = 6{,}278$, 26 INFERRED PHOTOGRAPHED PARENTS). IMAGE-RANDOM FOLDS DELIBERATELY LEAK RELATED VARIANTS; SCORES ARE EXPLORATORY.

Metric	Image-random CV	Parent-grouped CV
Pooled OOF macro-F1	0.8714	0.6879
Pooled OOF balanced acc.	0.8777	0.6901
Parent overlap per fold	26	0

TABLE II
FIRST FROZEN *inferred*-PHOTO-PARENT HOLDOUT (NEW MOBILENET MODEL; 1,086 AUGMENTED IMAGES, 5 PARENT CANDIDATES).

Metric	Value	Metric	Value
Accuracy	0.8020	Macro-F1	0.7995
Balanced acc.	0.8174	ROC-AUC	0.9135
Normal recall	0.8863	Faulty recall	0.7485
Normal n	422	Faulty n	664

Confusion matrix, rows/columns normal–faulty: $\left[\begin{smallmatrix} 374 & 48 \\ 167 & 497 \end{smallmatrix}\right]$.

TABLE III
VARIATION ACROSS THE FIVE *inferred* PHOTOGRAPHIC PARENTS IN THE FIRST FROZEN INTERNAL HOLDOUT (FIXED THRESHOLD).

Parent	N	Normal	Faulty	FP	FN	Acc.
AG	213	89	124	3	8	94.84
AK	333	76	257	13	94	67.87
AY	245	99	146	13	36	80.00
AZ	66	66	0	0	0	100.0
BB	229	92	137	19	29	79.04

C. Frozen grouped internal test

Table II shows the first, fixed 31-inferred-parent MobileNet holdout. Accuracy is 80.20%, macro-F1 is 0.799, balanced accuracy is 81.74%, and faulty recall is 74.85%. Thus 167 of 664 edited faulty samples are missed, even before authentic broken-stitch acquisition is tested. The held-out set represents just *five inferred photo parents*, so image-level sample size does not imply 1,086 statistically independent observations.

The five inferred parents are not equally difficult. For photo candidate `AK`, the model misses 94 of 257 faulty images, while `AG` has 8 missed faults among 124. Candidate `AZ` contains 66 normal images and *no* faulty example. This heterogeneous parent-level structure limits the interpretation of the aggregate score and prevents a claim of stable accuracy across fabrics or production conditions. The holdout is now inspected: any future model revision influenced by it requires new test photos.

D. First zero-adaptation StitchingNet transfer

Table IV concerns the two *earlier 182-family-trained* models only. Pretrained features substantially improve macro-F1 over the scratch CNN under domain shift, yet the MobileNet

TABLE IV
FIRST FROZEN EXTERNAL TEST: STITCHINGNET 3,637 IMAGES (*older 182-family models*, NO EXTERNAL FITTING).

Metric	Scratch CNN	Frozen MobileNet
Accuracy	0.3561	0.6646
Balanced accuracy	0.4969	0.6426
Macro-F1	0.3146	0.6354
Normal recall	0.0821	0.7073
Broken recall	0.9117	0.5779
Broken ROC-AUC	0.4821	0.6874

MobileNet confusion matrix $\left[\begin{smallmatrix}1723 & 713\\ 507 & 694\end{smallmatrix}\right]$; rows/columns normal–broken.

misses 507 of 1,201 genuine broken stitches. Its 66.46% accuracy is even slightly below the 66.98% achieved by classifying every external image as normal. Macro-F1 and class-wise recall, not accuracy alone, better describe operational usefulness. We do *not* attribute the StitchingNet score to the later 31-photo-parent-trained checkpoint.

E. Class imbalance and evaluation-unit sensitivity

A trivial all-normal classifier scores $2{,}436/3{,}637 = 66.98\%$ accuracy on the external binary StitchingNet subset, compared with 66.46% for the older MobileNet. Thus a nominally high accuracy can hide a weak defect-screening strategy. Our comparison emphasizes both class recalls and macro-F1. For the 31-parent internal test, there are only five inferred independent candidate sources and one parent (AZ) contains no faulty example. The apparent $N = 1{,}086$ is therefore inappropriate for image-independent inferential confidence intervals. Reporting per-parent confusion counts is preferable to overstated narrow uncertainty bounds. The next step is not post-hoc tuning of inspected tests but prospective image collection with garment-level grouping and source rights.

F. Implications for garment inspection

Both evaluations show the importance of making *false negatives* visible. Internal synthetic holdout performance cannot establish production deployment quality; external real broken-stitch recall of 57.79% is also insufficient for unattended inspection. The case-specific contribution is transparent evidence accounting: family-level overlap in the original partitions, a development-only controlled split-protocol comparison, an inferred-parent holdout and an external model comparison appropriately separated from later training. This is a methodologically informative negative result, not a claim that the technique outperforms recent defect-localization systems.

G. Distinct error modes and aggregation effects

The small number of source-photo candidates changes how these performance measures should be read. Consider two hypothetical operating points with the same overall accuracy but different sensitivities: one may minimize unnecessary reinspection of intact garments, while the other misses a greater fraction of structurally broken seams. At the fixed 0.5 threshold in our initial internal holdout, the model flags 48 of 422 normal images as faulty and marks 167 of 664 Photoshop-faulty images as normal. The latter error is more consequential for unattended quality inspection. There is no evidence that a different threshold would improve the real-fault trade-off on independent production images; threshold selection on the now-inspected test would invalidate its role as a frozen holdout.

An independent-seeming count of 1,086 augmented images also understates sampling correlation. Every held-out record is associated with one of just five inferred photographed sources, many derivatives of a source appearing repeatedly. Therefore an image-wise bootstrap or conventional binomial interval would likely overstate precision. Even computing a parent-level interval is difficult with only five clusters, including one all-normal source. We instead report the full per-parent confusion counts and refrain from estimating a narrow population accuracy interval. A future prospective experiment should sample many physically different garments, seams and camera acquisition sessions and should keep images of one garment together.

The ROC-AUC of 0.914 in the grouped internal set characterizes ranking over synthetic/augmented examples, not detection at the operational threshold or performance on naturally broken seams. It cannot negate the observed 25.15% false-negative proportion for the faulty internal class. The older MobileNet's external ROC-AUC of 0.687 likewise does not negate its 42.21% broken-seam miss proportion. The original 94.91% historical validation number is not directly comparable with either of these values: it concerns a different and not fully reconstructed configuration, evaluation history, and possibly different partitions.

H. Reproducibility controls and unperformed claims

To preserve the two distinct model lineages, experiment outputs were archived separately. The source ZIP is identified by SHA-256 `e7aed1355a712c14bb...`; the inferred-parent MobileNet holdout checkpoint by `02fa682d0648aa2521...`; and its split manifest by `769d10dc6f6651f2...`. The frozen evaluation verifies the full digests, class mapping and fixed threshold before reading the holdout. The training workflow for the second MobileNet never loads StitchingNet; consequently there is *no* independent external performance estimate for that second model. Meanwhile the external evaluation used only models frozen *before* StitchingNet performance was examined. Once a benchmark has been inspected, exploratory reuse cannot be described as another untouched test.

I. Evidence still required for prospective deployment

A new real-world evaluation must record the true physical-seam or garment identifier, original photographic file, camera, lighting, textile/thread combination, whether the defect was physically present, and independent expert review of each defect label. Photoshop-created positives should be marked separately rather than mixed with authentic positives. At least two reviewers should resolve label disputes blind to predicted class, with image and source records retained subject to owner

consent. This would allow genuinely unseen source groups, per-fabric sensitivity estimates, prevalence-aware precision, confidence intervals clustered by garment, and a credible assessment of whether broken-stitch recall is high enough for a human-in-the-loop production setting. Until such data exist, the present scores are evidence about reproducibility and failure modes, not certification of a factory inspection product.

VI. Limitations and Conclusion

Original unedited photographs, Photoshop edit histories, and per-image ownership/source logs are missing. The syntactic 151-to-31 parent mapping is supported by filenames and visual similarity but remains unconfirmed by the original image creators. The number of truly distinct garment sources may be smaller than 31. The 7,364 readable images contain synthetic defects and transformations; only five inferred parents entered the internal test. Results are sensitive to parent composition, labels, domain shift, camera setup and a single experiment seed. The original 2022 record also contains inconsistent augmented-image counts and does not preserve the run needed to reproduce its published validation score. We have not performed a new independently sourced real broken-seam test for the later 31-photo-parent checkpoint; public derivative releases of StitchingNet are not a second independent dataset.

In conclusion, this *case-specific* audit demonstrates that *all 182 augmentation families* in the recovered source ZIP cross original partitions. A controlled development-only partition diagnostic on the same 6,278 records shows an observed 18.35-point macro-F1 difference between image-random and inferred-parent-grouped cross-validation, consistent with considerable partition sensitivity but not a causal re-estimate of historical accuracy. After inferred-photo-parent grouping, a pretrained MobileNet classifier scores 0.799 macro-F1 on a small augmented-image holdout, whereas an *earlier* pretrained model yields 0.635 macro-F1 with 57.79% defect recall on previously unseen real StitchingNet images. These experiments demonstrate the importance of provenance-aware splitting and honest external metrics, but they do not validate photo-source independence or operational reliability. These observations are not a novel solution to augmentation leakage, nor a causal remeasurement of the historical validation claim. The next necessary evidence is human confirmation of original photo/edit lineage and a new, separately acquired, expertly labelled collection of genuinely broken sewn seams.

Generative-AI Use Disclosure

ChatGPT assisted with literature exploration, research software and experiment workflow design, manuscript drafting and LaTeX editing. Numerical metrics reproduced here were generated by archived, executable GitHub Actions runs rather than invented. The human authors are responsible for checking source provenance, interpretations, originality, citations, data permissions and conference rules before submission.

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